

```
0.          /* Bitpos */
complain_overflow_dont, /* Complain_on_overflow */
bfd_elf_generic_reloc, /* Special_function */
"R_CR16_NONE", /* Name */
FALSE, /* Partial_inplace */
0. /* Src_mask */
0. /* Dst_mask */
FALSE, /* PCREL_offset */
```

3. Define the macros below with settings specific to the target:

```
#define TARGET_LITTLE_SYM
#define TARGET_LITTLE_NAME

#define ELF_ARCH
#define ELF_MACHINE_CODE
#define ELF_MAXPAGESIZE
#define elf_symbol_leading_char

#define elf_info_to_howto
#define elf_info_to_howto_rel

#define elf_backend_relocate_section
#define elf_backend_gc_mach_hook
#define elf_backend_gc_sweep_hook
#define elf_backend_can_gc_sections
#define elf_backend_rela_normal
#define elf_backend_check_relocs
#define elf_backend_final_write_processing
#define elf_backend_object_p
#define elf_backend_create_dynamic_sections
#define elf_backend_adjust_dynamic_symbol
#define elf_backend_size_dynamic_sections
#define elf_backend_emit_section_dynsym
#define elf_backend_finish_dynamic_sections
#define elf_backend_reloc_type_class
#define elf_backend_want_got_plt
#define elf_backend_plt_reloc
#define elf_backend_want_plt_sym
#define elf_backend_got_header_size

#define bfd_elf32_bfd_reloc_type_lookup
#define bfd_elf32_bfd_reloc_name_lookup
#define bfd_elf32_bfd_reloc_section
#define bfd_elf32_bfd_get_relocated_section_contents
#define bfd_elf32_bfd_merge_private_bfd_data
#define bfd_elf32_bfd_link_hash_table_create
#define bfd_elf32_bfd_link_hash_table_free
```

In addition, the files *archures.c*, *configbfd*, *Makefile.am* and *target.c* in the *bfd* directory should be updated with the necessary target-specific changes.

Opcodes

The *opcodes* directory should have at least two target-specific files—one to assemble the target instructions and the other to

disassemble them. The file names are:

1. *<target>-opc.c* and *<target>-opc.h* (header files are optional)
 2. *<target>-disc.c* and *<target>-disc.h* (header files are optional)
- The *<target>-disc.c* files include code to print disassembled instructions, and also code for matching opcodes and operands appropriately.

The *configure.in*, *disassemble.c* and *Makefile.am* files in the *opcodes* directory also need to have target-specific changes made to them.

Include

The *include* directory contains target-specific header files, usually in a file format-specific subdirectory. *Opcode* information is normally held in a target-specific file in the *opcode* subdirectory. For example:

```
include/elf/cr16.h
include/opcode/cr16.h
```

Binutils

This directory doesn't require any new target-specific files. But the *configure.tgt*, *Makefile.am* and *readelf.c* files should be updated with any target-specific information needed.

Gas

The *gas/config* subdirectory contains the target-specific files for the assembler. The file names are *tc-<target>.c* and *tc-<target>.h*.

The above files should include:

- Sizes of (i.e., macros defines):
 - Registers
 - Instructions (i.e., maximum size)
 - Operands
- Operand error types
- Comment character used in assembly code
- Comment character used in assembly code line
- Line separator
- Defining the target-specific
 - multi-character options, if any.
 - Process machine-dependent command line options in the *md_parse_option* function
 - Include machine-dependent usage-output in the *md_show_usage* function
- A redefinition of the assemble directive using *md_pseudo_table*
- Functions for getting registers along with type, size, and other information
- *md_begin* function used to initialise/set-up all the tables, etc, that are machine-dependent items of the assembler
- Parse functions:
 - *parse_insn*
 - *parse_operands*
 - *parse_operand*
- Print functions:
 - *print_insn*